

Q4) a) What is load prediction? What is its necessity? What are the different methods of load prediction? [7]

b) A run-of-stream station with an installed capacity of 12,000 kW operates at 30% weekly load factor when it serves as a peak load station and its entire capacity is firm capacity. What should be the lowest discharge in the stream so that the station may serve as the base load station? It is given that the plant efficiency is 75% when working under a head of 25m. Also calculate the daily load factor of the plant when the discharge in the stream rises to 20 cumce. [8]

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Total No. of Questions : 4]

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B.E. (Civil) (Insem)

HYDROPOWER ENGINEERING

(2019 Pattern) (Semester - VIII) (Elective - V) (401013E)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data, if necessary.*
- 4) *Neat diagrams must be drawn wherever necessary.*

Q1) a) Explain with neat sketch the elements of hydropower generation scheme. **[7]**

b) What is need of energy? Explain renewable and nonrenewable energy sources with their advantages and disadvantages. **[8]**

OR

Q2) a) Explain in detail classification of hydropower plants. **[7]**

b) Explain with neat sketch the main divisions of surface power house. **[8]**

Q3) a) What is Tariff? Explain different Methods Tariff in detail. **[7]**

b) What a detail note on :

- i) Run of river plant with and without pondage
- ii) Pumped Storage plant **[8]**

OR

P.T.O.